

PAN Card Layout Analysis for OCR

1 PAN Card Layout Variations

The Permanent Account Number (PAN) card is a unique identification document issued by the Income Tax Department of India. Over the years, PAN cards have evolved from legacy physical cards to digitally issued e-PAN documents with improved security features. These layout variations introduce differences in document structure, typography, security patterns and image quality that directly influence Optical Character Recognition (OCR) performance.

1.1 Official e-PAN



The e-PAN is a digitally generated document containing the PAN card, verification QR code and issuance details. Since it is generated electronically, it provides high-quality text and serves as a reference layout for OCR benchmarking.

1.2 Legacy PAN Card

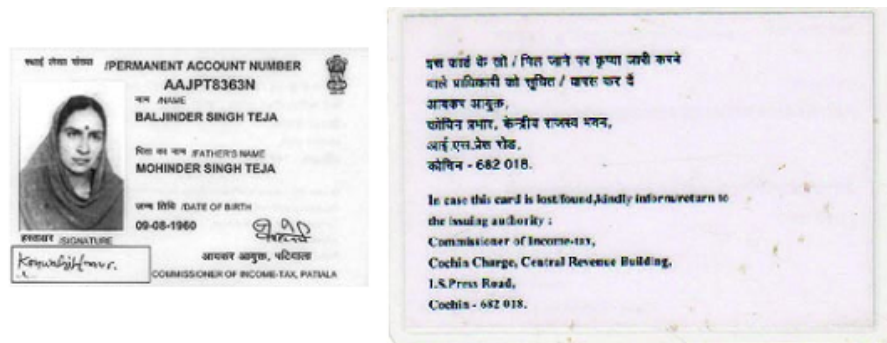


Figure 2: Legacy PAN card layout (front and back).

Older PAN cards differ considerably from current versions in typography, background design and information placement. OCR systems frequently encounter scanned copies of these documents during archival digitization.

1.3 Current PAN Card



Figure 3: Current PAN card (front side).

The current PAN card incorporates enhanced security features, a redesigned background and standardized field locations. It is the most commonly used physical PAN card for identity verification.

1.4 Current PAN Card (Reverse Side)

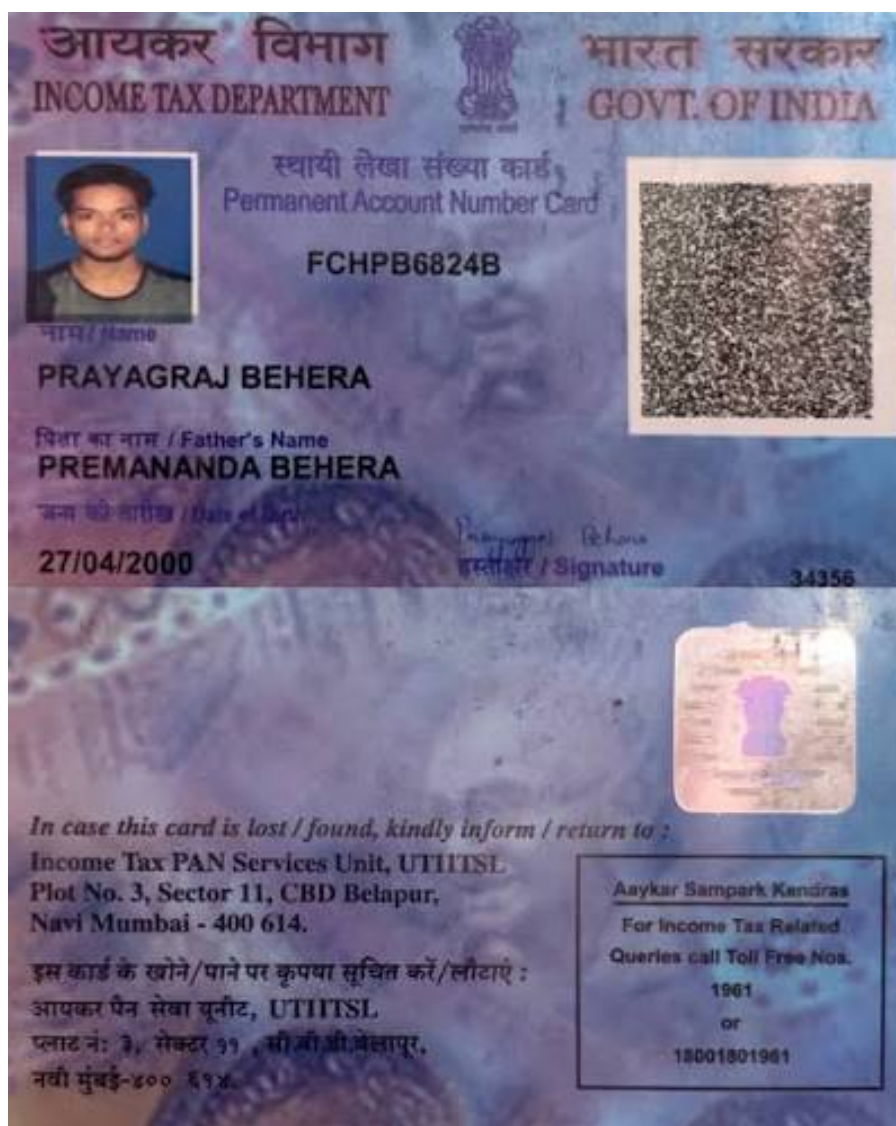


Figure 4: Reverse side of the current PAN card.

The reverse side contains important instructions and security information. Although it contains comparatively less structured data, OCR systems must still extract printed text accurately for document verification.

2 Document Structure

The PAN card follows a standardized layout containing identity information, issuing authority details and security features.

Table 1: Primary PAN Card Identifiers

Identifier	Description
PAN Number	Unique 10-character alphanumeric identifier
Name	Cardholder name
Father's Name	Parent's name
Date of Birth	Date of birth
Photograph	Identity photograph
Signature	Cardholder signature
QR Code	Digital verification (new layouts/e-PAN)
Income Tax Department	Issuing authority
Government of India Logo	National emblem

3 Identifiers and OCR Keywords

The OCR system utilizes textual anchors to identify the document type and accurately locate important fields.

- Government of India
- भारत सरकार
- Income Tax Department
- आयकर विभाग
- Permanent Account Number
- PAN
- PAN Card
- Name
- Father's Name
- Date of Birth
- DOB
- Signature
- Photograph
- QR Code
- PAN Number
- स्थायी खाता संख्या

4 OCR Challenges

PAN cards present several OCR challenges because of variations in document generation, printing quality and capture conditions.

- Multiple PAN card layouts
- Legacy and current document formats
- Small font sizes
- Security background patterns
- Signature extraction
- Photograph localization
- QR code detection

- Low-resolution scanned copies
- Shadows and reflections
- Perspective distortion
- Compression artifacts

Table 2: Relative OCR Difficulty Across PAN Card Layouts

Layout	OCR Difficulty
Official e-PAN PDF	Low
Legacy PAN Card	Medium
Current PAN Card	Medium
Scanned/Photographed PAN Card	High